

Phylogenetics

Introduction to comparative
methods

RL-V3 MPP

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Phylogenetic comparative methods

Comparative methods

The distribution of traits among species is the result of evolution

Traits can be **discrete** or **continuous**

Comparative methods are approaches used to study trait evolution

Why do we need to take phylogeny into account when we study trait evolution?

CONTINUOUS

measured data, can have ∞ values within possible range.



I AM 3.1" TALL
I WEIGH 34.16 grams

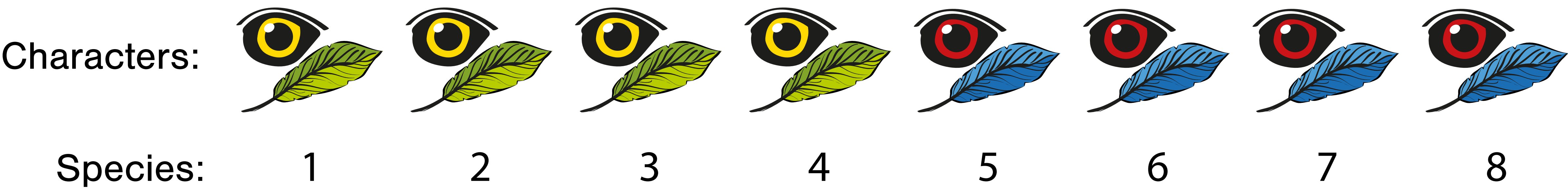
DISCRETE

OBSERVATIONS can only exist at limited values, often COUNTS.



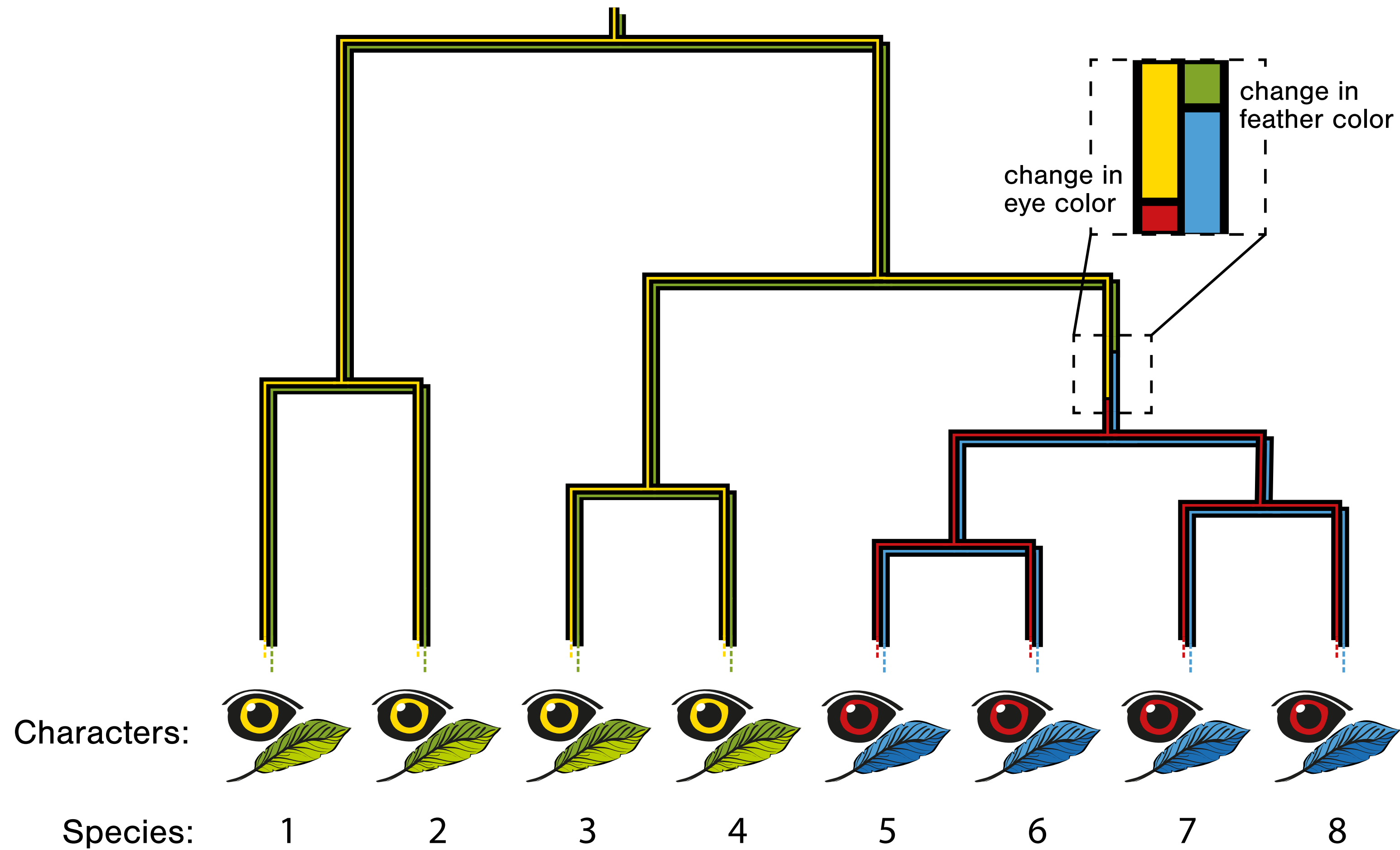
I HAVE 8 LEGS
and
4 SPOTS!

@allison-horst

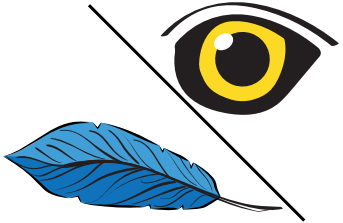


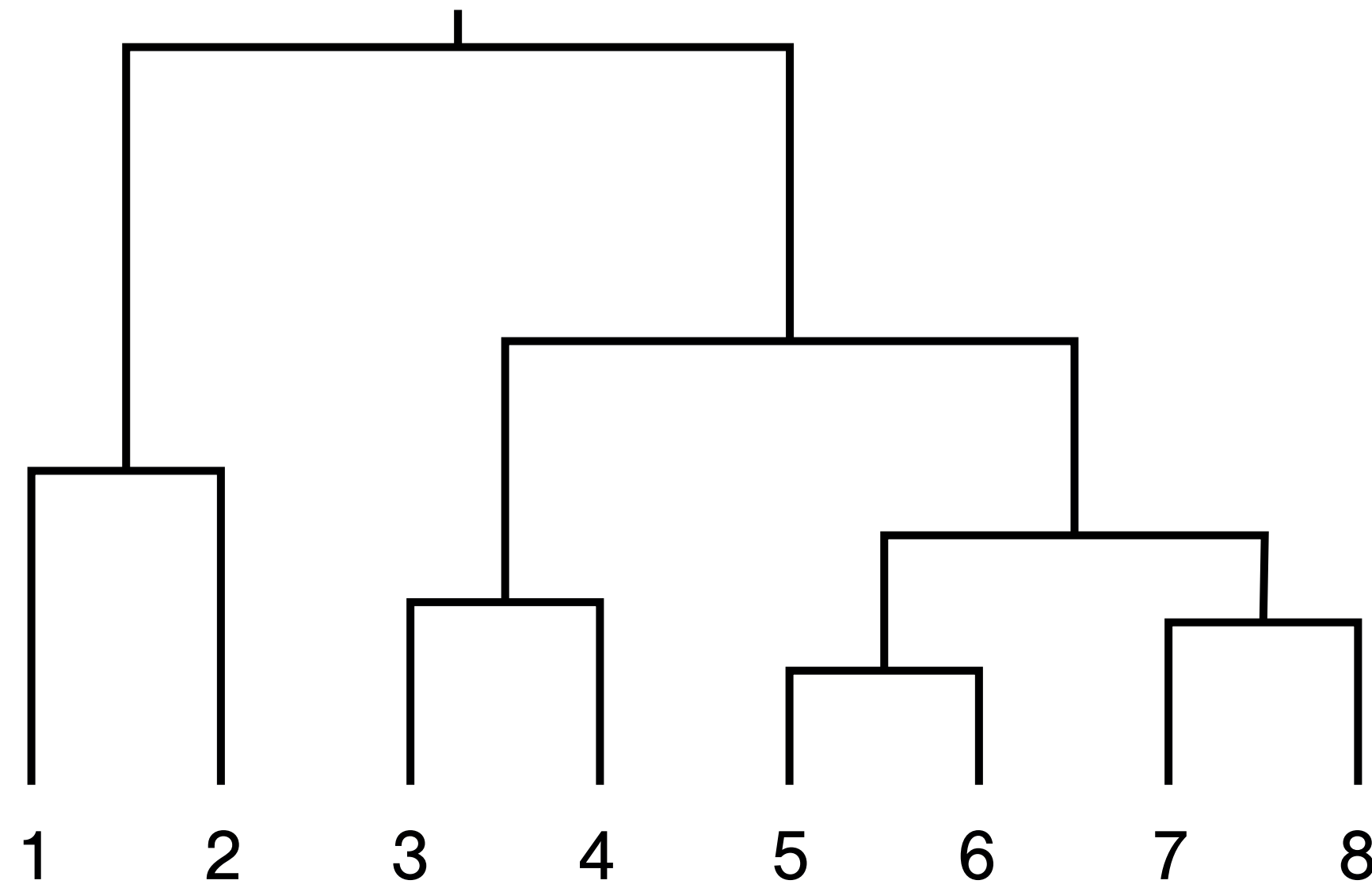
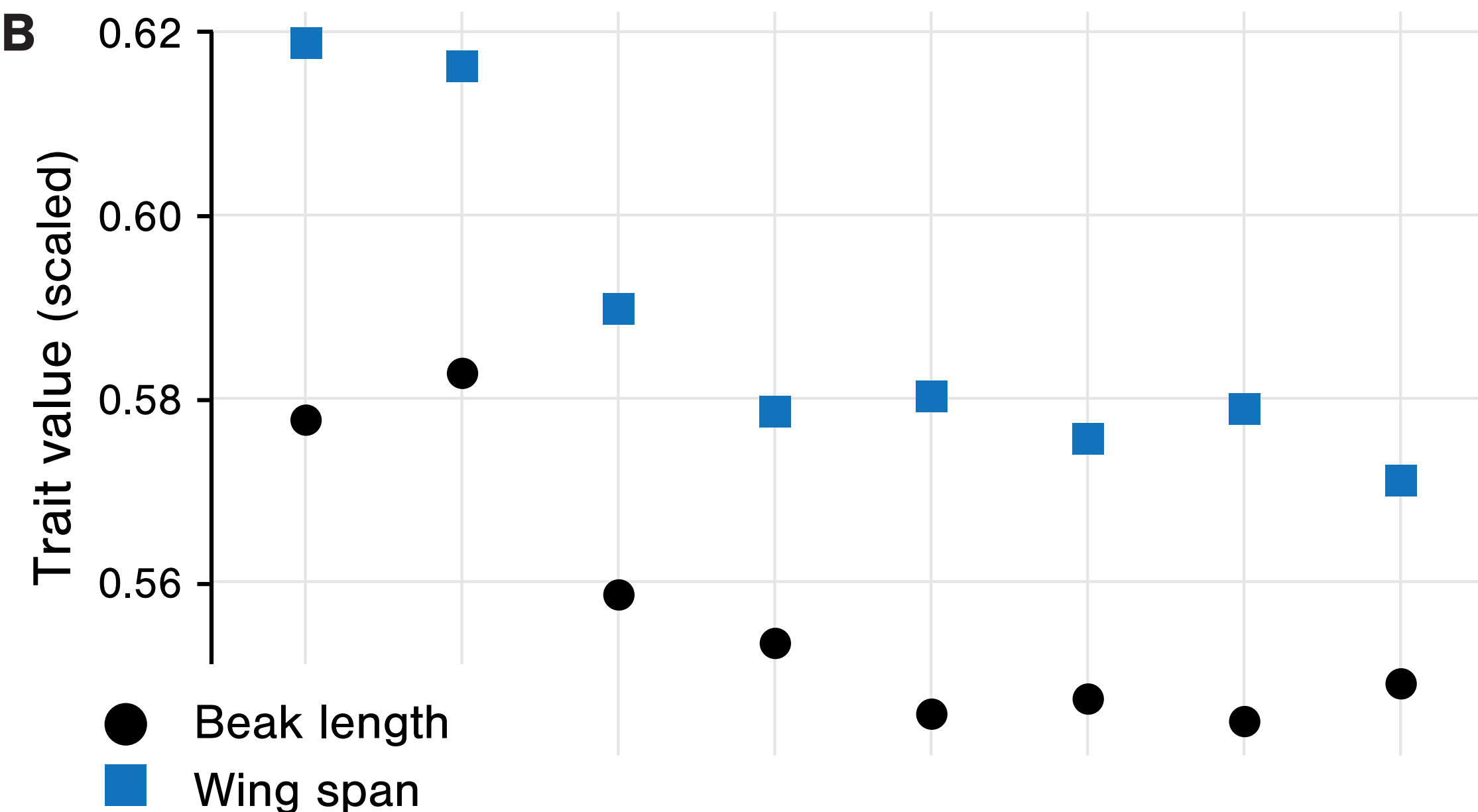
Contingency table

			Totals
	4	0	4
	0	4	4
Totals	4	4	8



Contingency table
(changes per branch)

	Change	No change	Totals
Change	1	0	1
No change	0	13	13
Totals	1	13	14

A**B**

The tree shows bird species, along with continuous wing span and beak length traits for each one

Note the trait values highly depend on which of the two main clades of a given species belongs to!

Phylogenetic comparative methods (PCMs)

A large family of statistical approaches used to study trait evolution that take into account phylogenetic non-independence

The input is usually a tree, combined with trait data

Examples: tests for phylogenetic signal, phylogenetic PCA (morphometrics), ancestral state reconstruction, evolutionary model fitting analysis (tempo and mode), multi-type diversification models (see previous weeks), many more

Can you think of any questions you would ask about trait evolution?

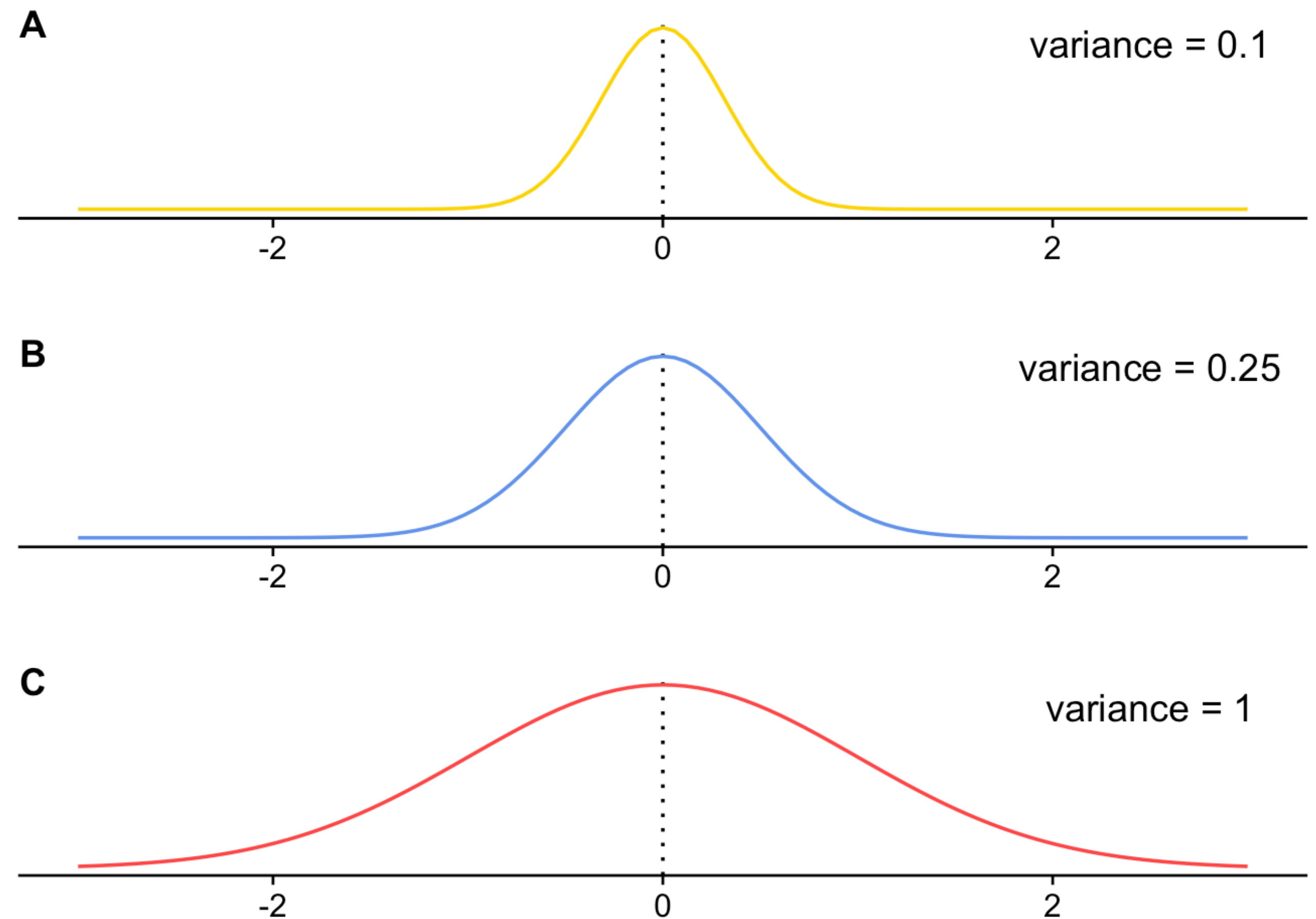
Models of continuous trait evolution

Brownian motion (BM)

The amount a trait changes over time is drawn from a normal distribution with a **mean of 0** and a **variance of σ^2**

Positive numbers cause the trait to increase

Negative numbers cause the trait to decrease



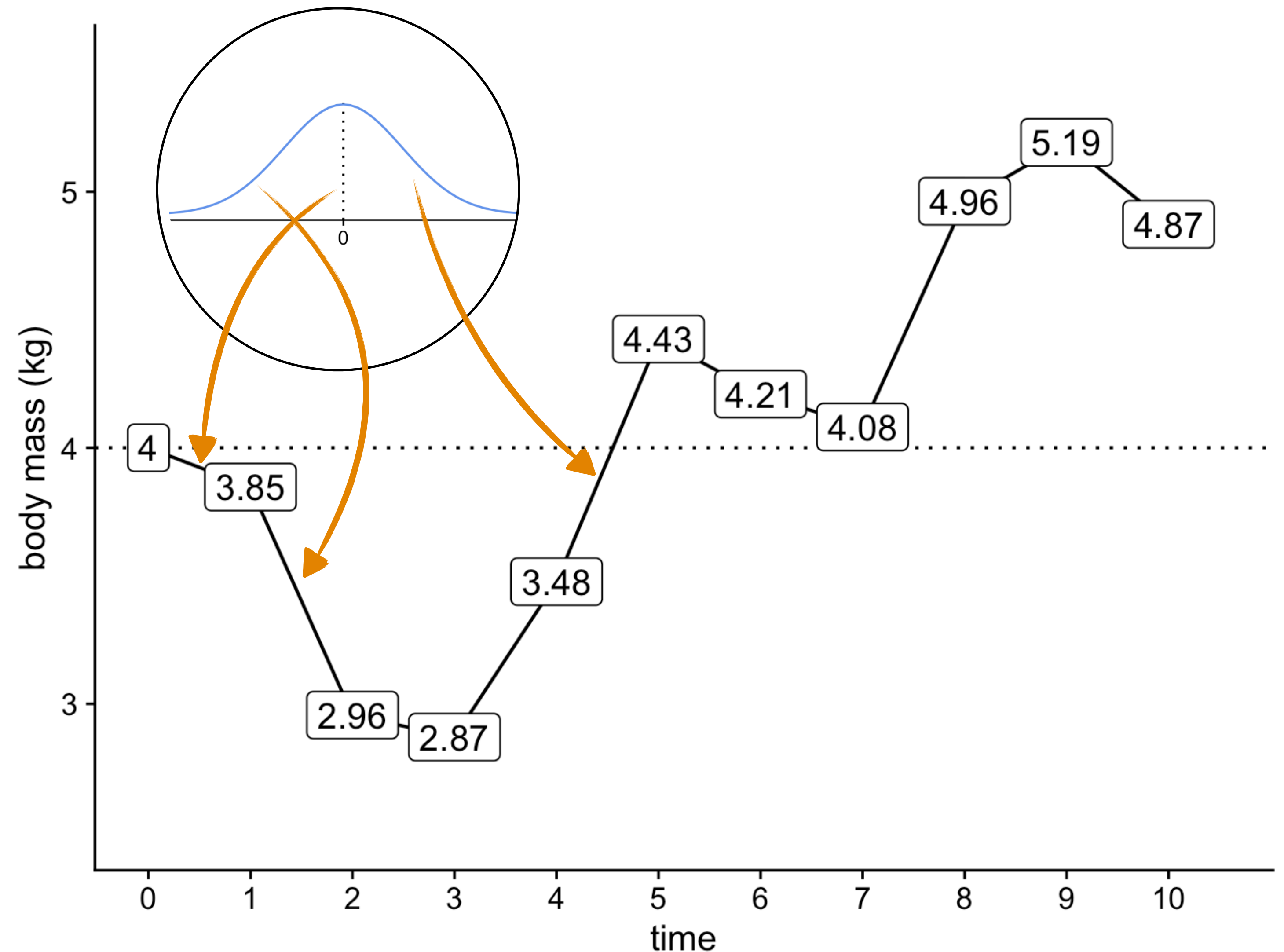
Brownian motion

Also known as a “random walk” model

A single model parameter: σ^2

Equivalent to the rate of evolution, larger σ^2 = faster evolution

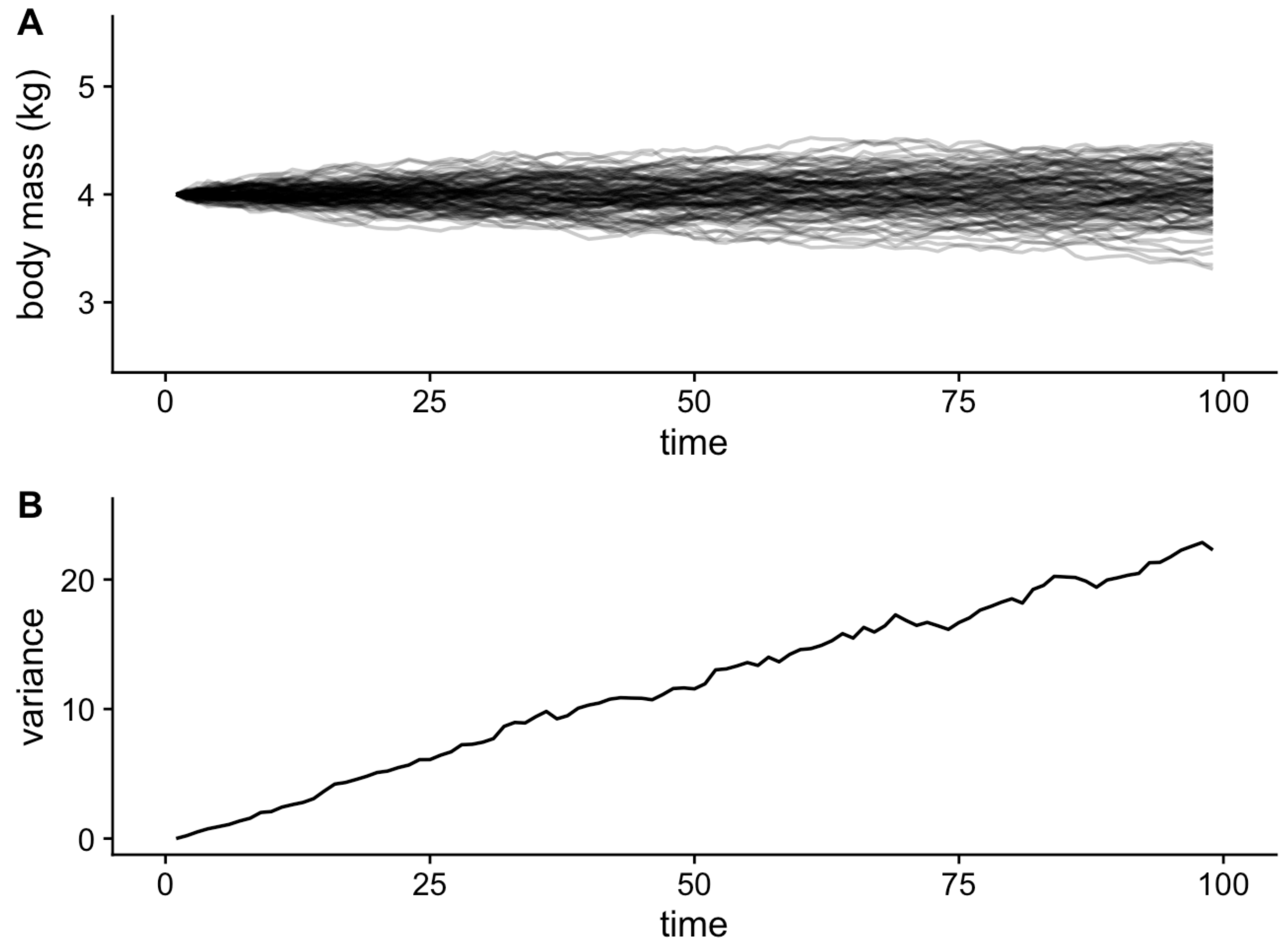
Figure shows a simple example of body mass change through time, with starting value = 4 and $\sigma^2 = 0.25$



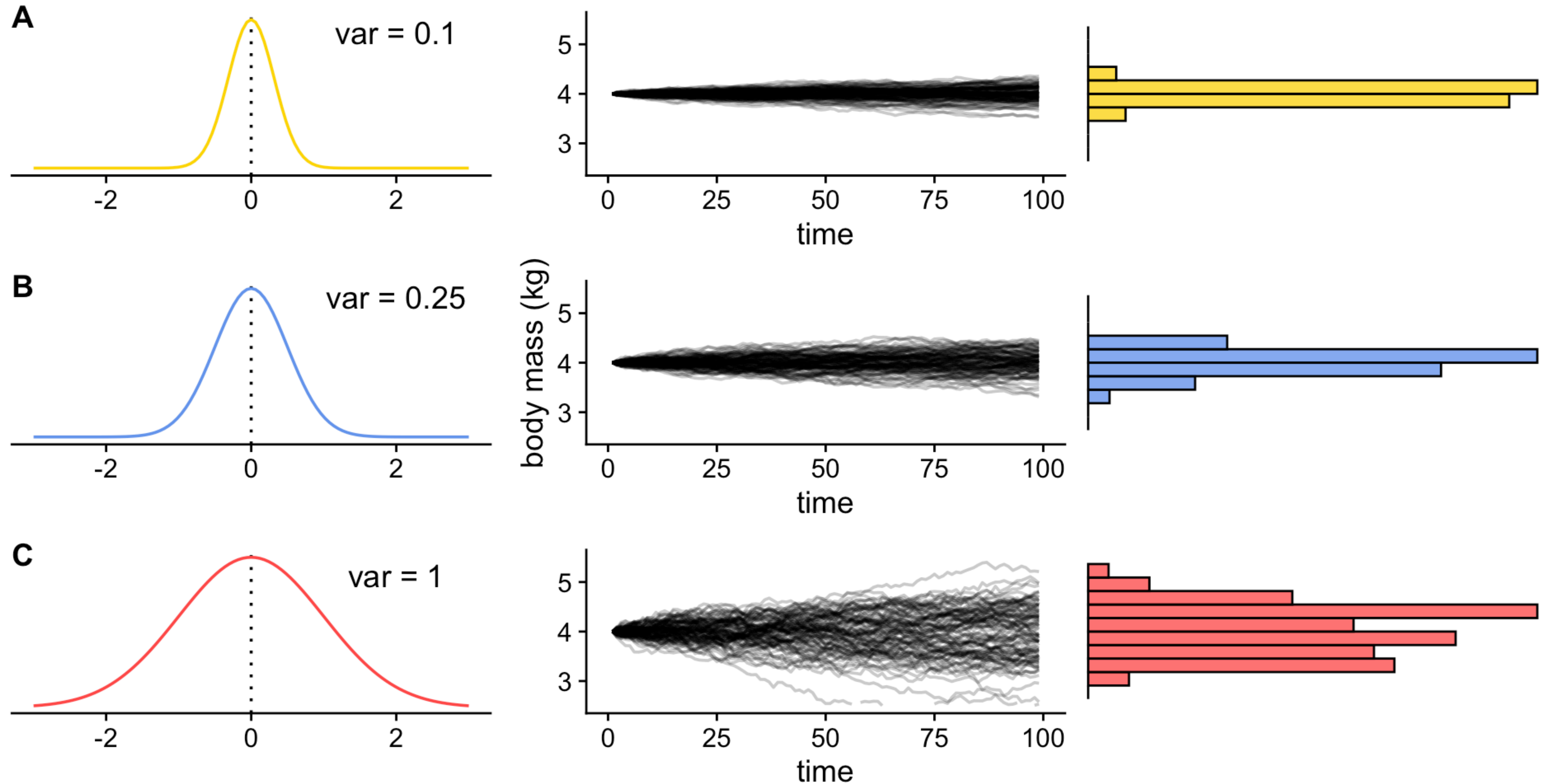
Brownian motion

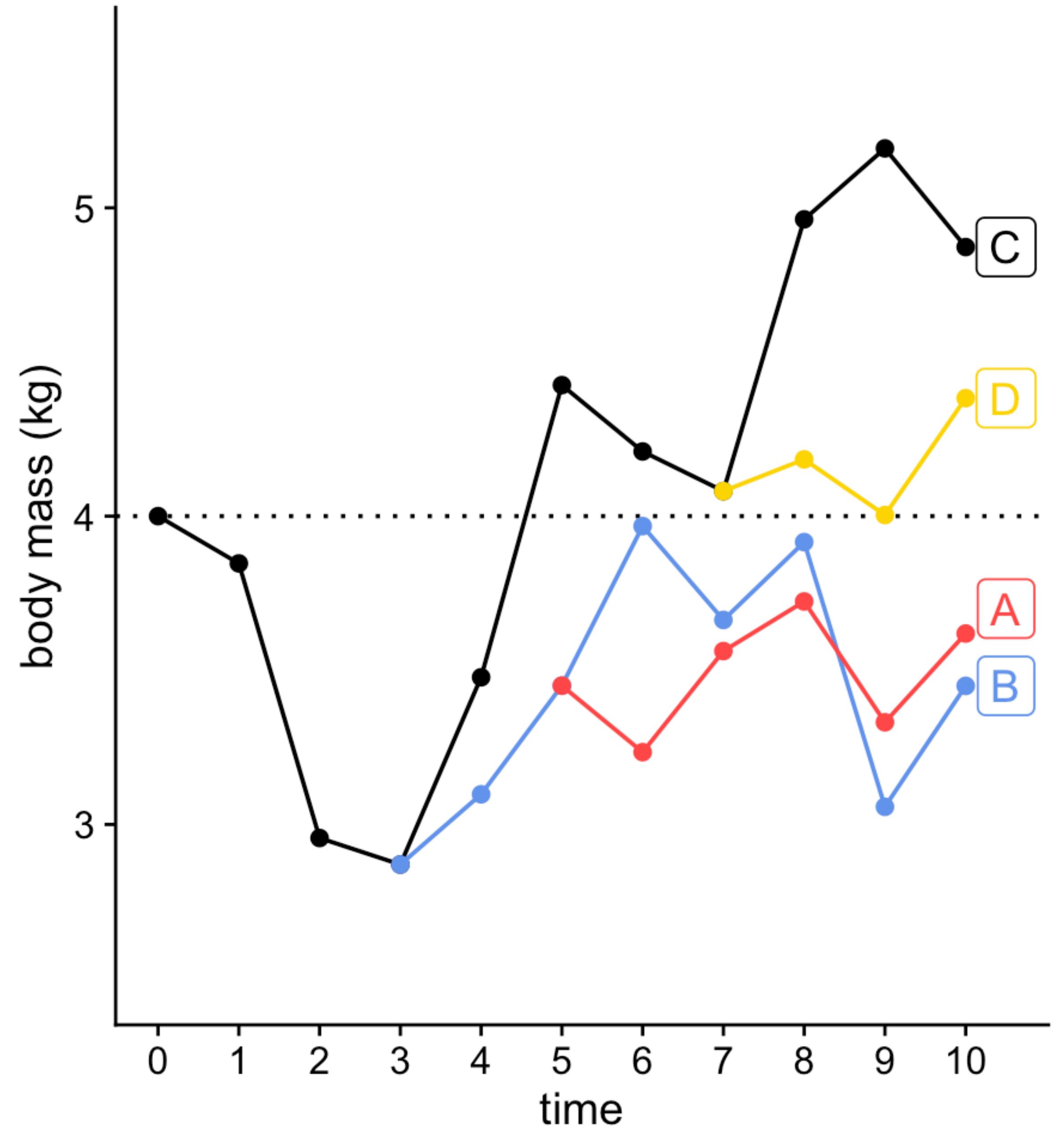
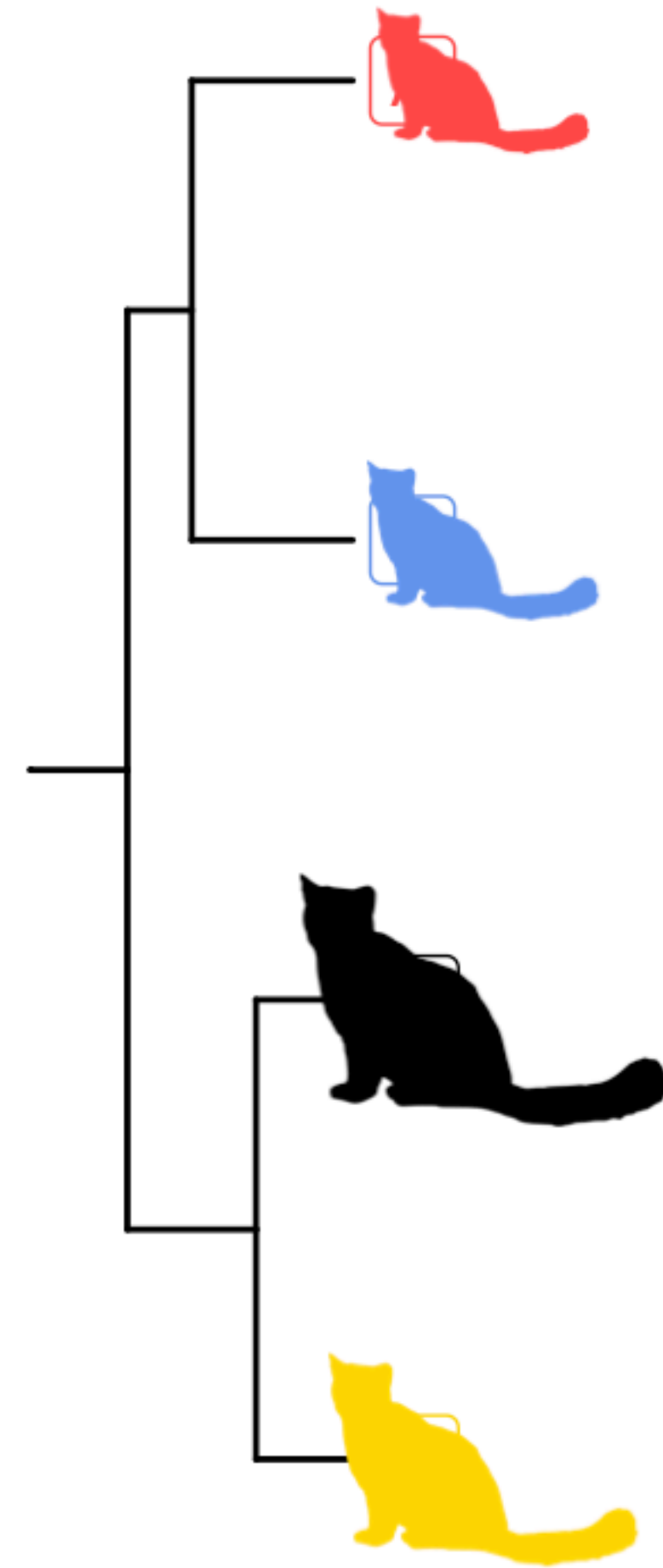
100 examples of body mass change through time under a BM model, with starting value = 4 and $\sigma^2 = 0.25$

The plot below shows the increase in the variance in body mass through time

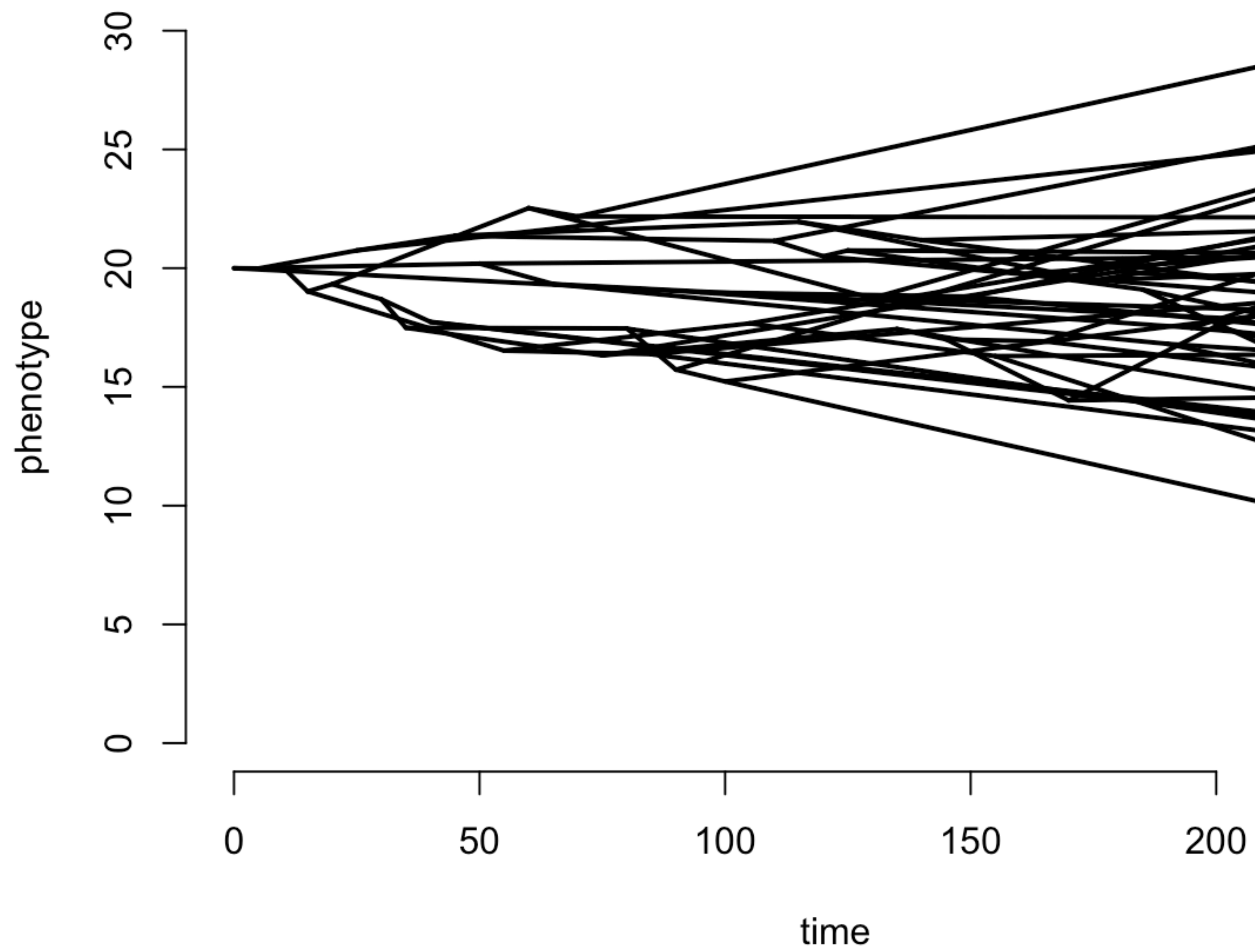


3 examples of body mass change under BM with the same starting value and 3 different values of σ^2





We can model BM along a tree



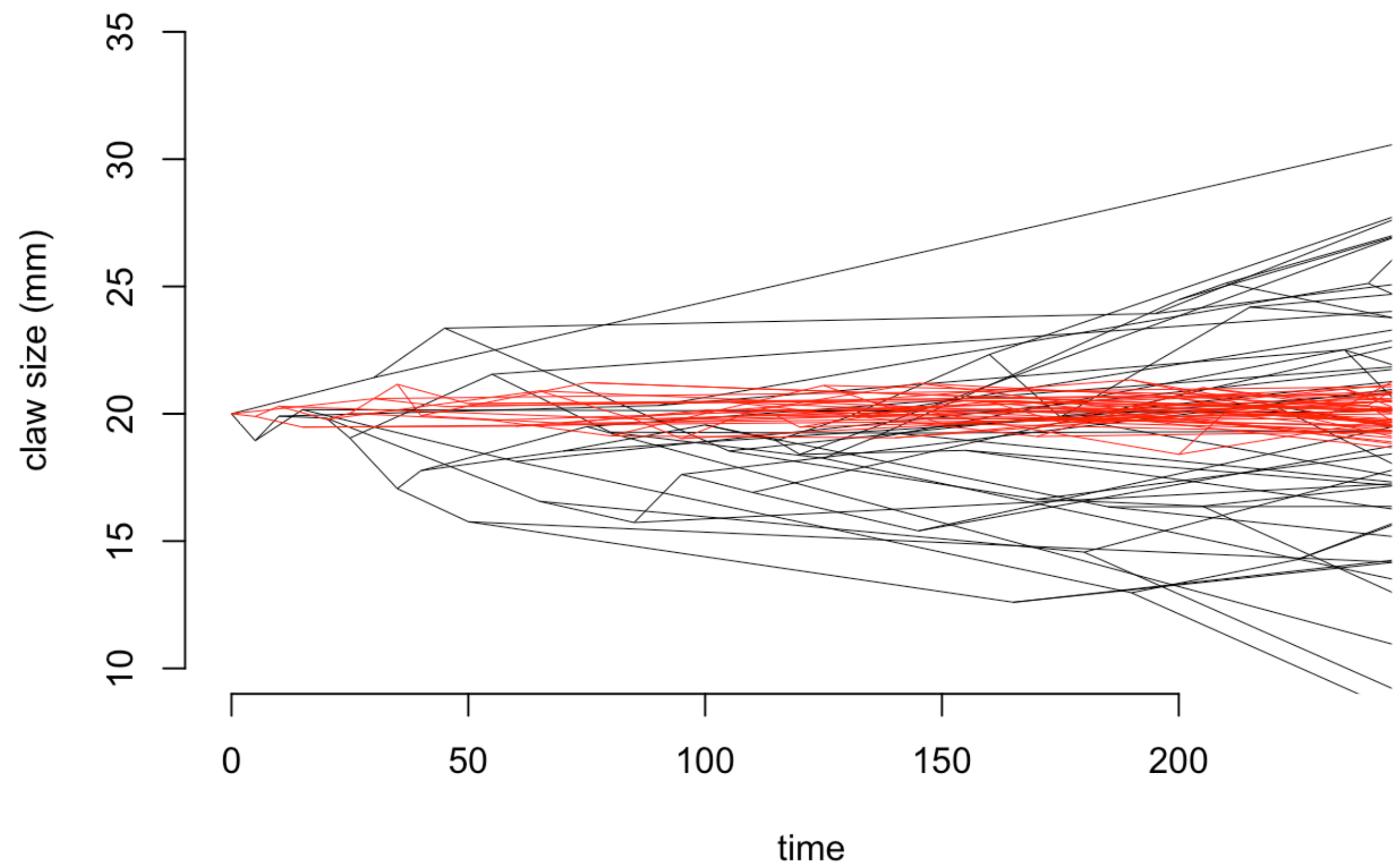
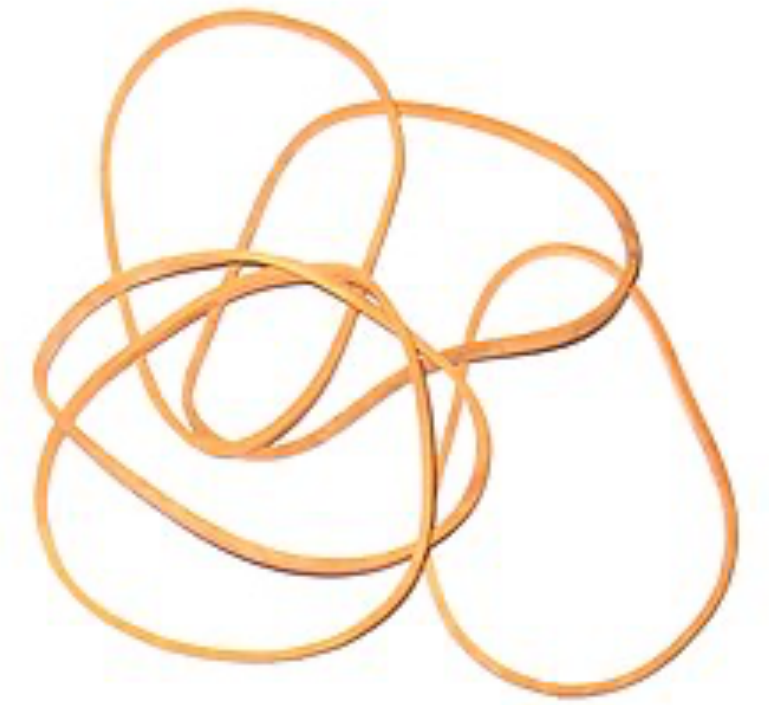
Ornstein-Uhlenbeck

Two model parameters:

the rate parameter σ^2

the rubber band parameter α

Larger α means the trait evolves closer to the mean, α close to 0 is equivalent to a random walk



PCMs require time trees

In theory, we could infer the tree and PCM parameters at the same time (and sometimes we do), but this is computationally tricky — usually we estimate the tree (topology and times) first

Make sure the method is appropriate for your data and don't over interpret results

Cooper et al. [2016](#)

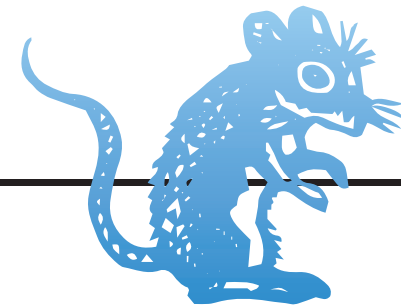
Fossils help! Simulation show including fossils improves our ability to identify the correct model of trait evolution *Slater et al.* [2012](#), *Gearthy et al.* [2026](#)

Typical workflow

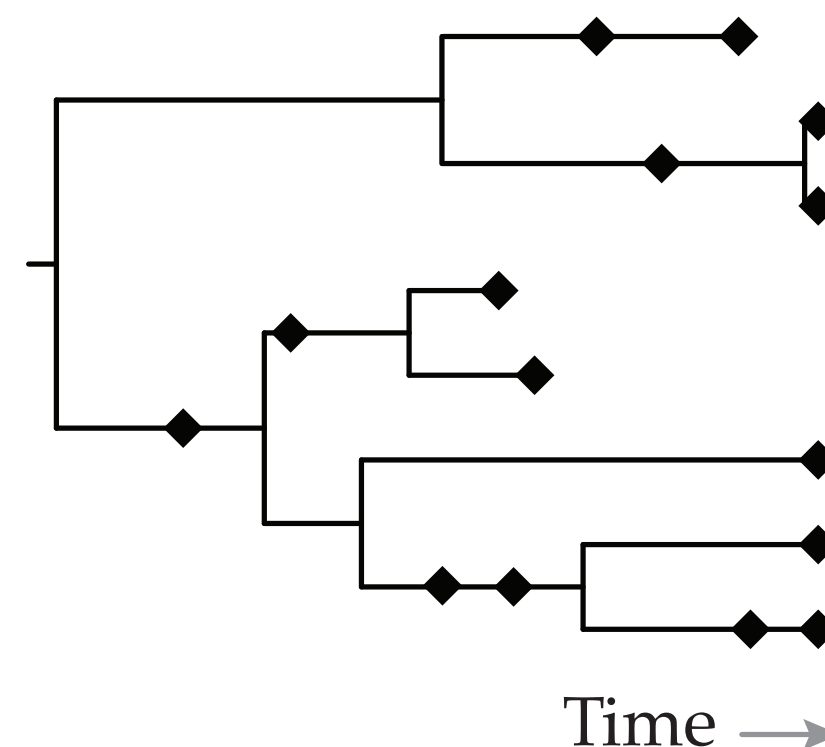
A. Phylogenetic data collection

<i>Inia</i>	0 Ma
A T G C	0 1 0 1
<i>Tursiops</i>	5.3 – 0 Ma
A T G C	0 1 0 1
<i>Isthminia</i> †	5 – 2 Ma
	0 1 0 1
<i>Eurhinodelphis</i> †	12 – 9 Ma

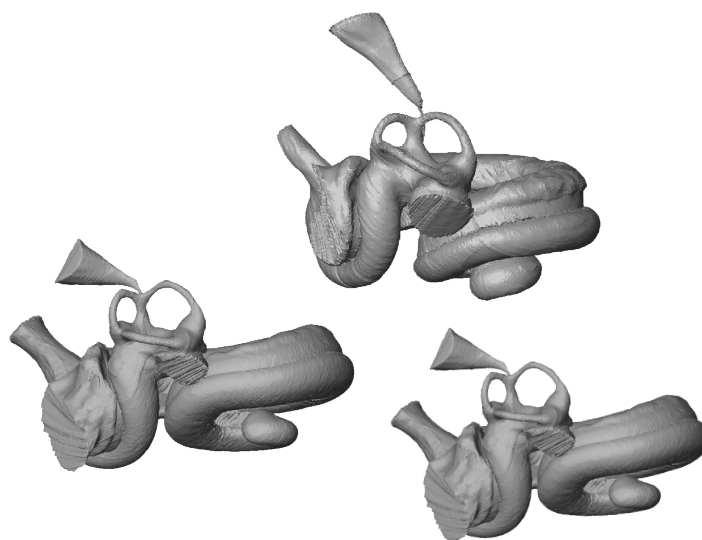
B. Analysis in BEAST2



C. Output part 1



D. CT scanning



E. Trait data collection

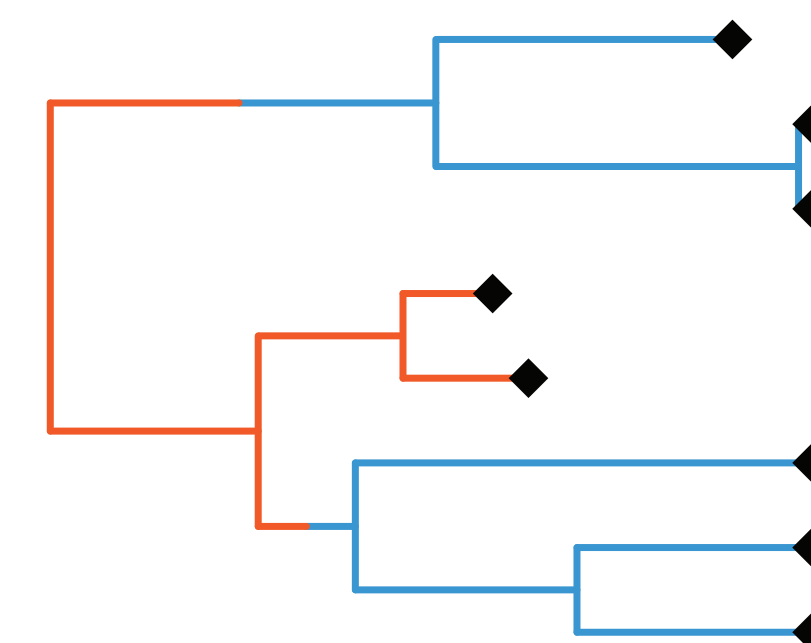
$$\begin{pmatrix} X_{11} & X_{21} & X_{31} & \dots & X_{N1} \\ X_{12} & X_{22} & X_{32} & \dots & X_{N2} \\ \dots & \dots & \dots & \dots & \dots \\ X_{1k} & X_{2k} & X_{3k} & \dots & X_{Nk} \end{pmatrix}$$

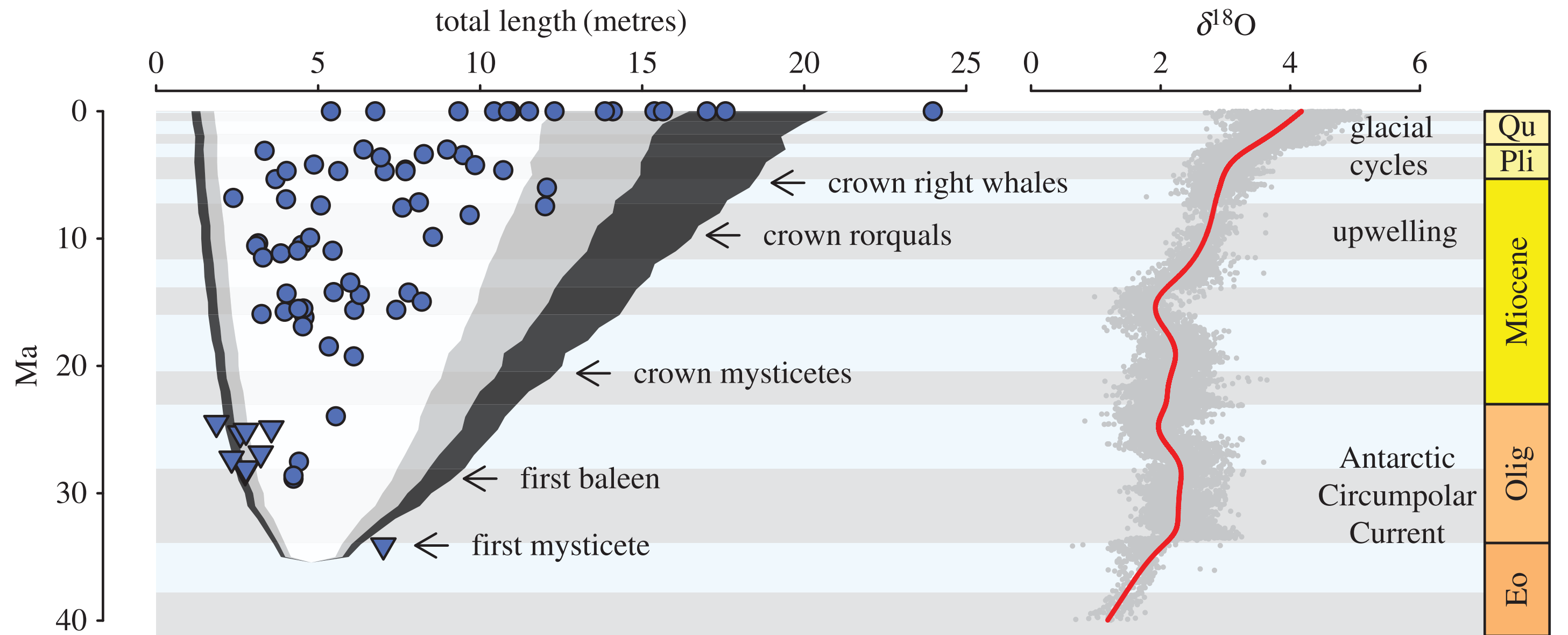
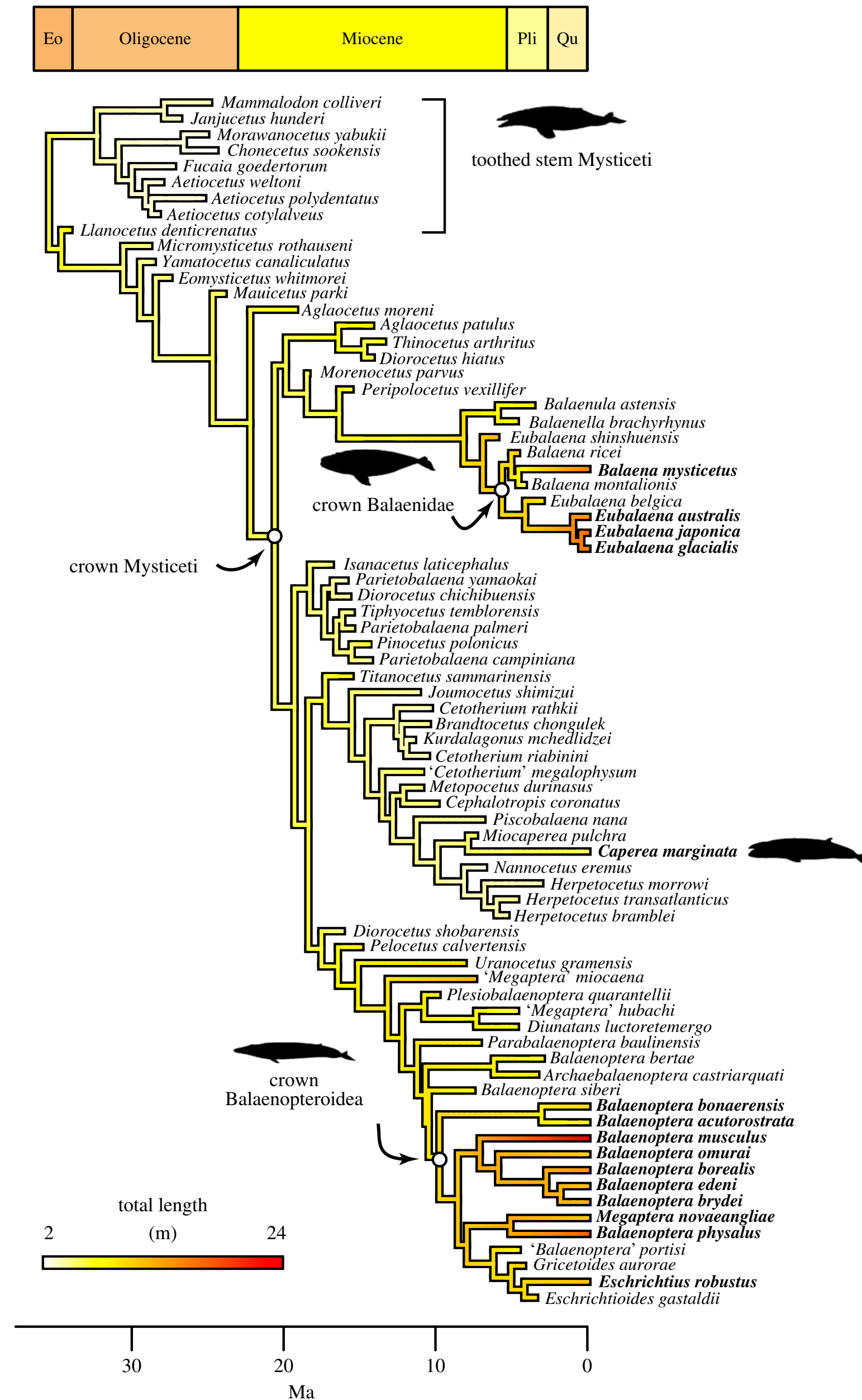
matrix of N taxa & k traits

F. Analysis in PCMBase



G. Output part 2





Example: gigantism in baleen whales

Linked to ocean dynamics in the Plio-Pleistocene

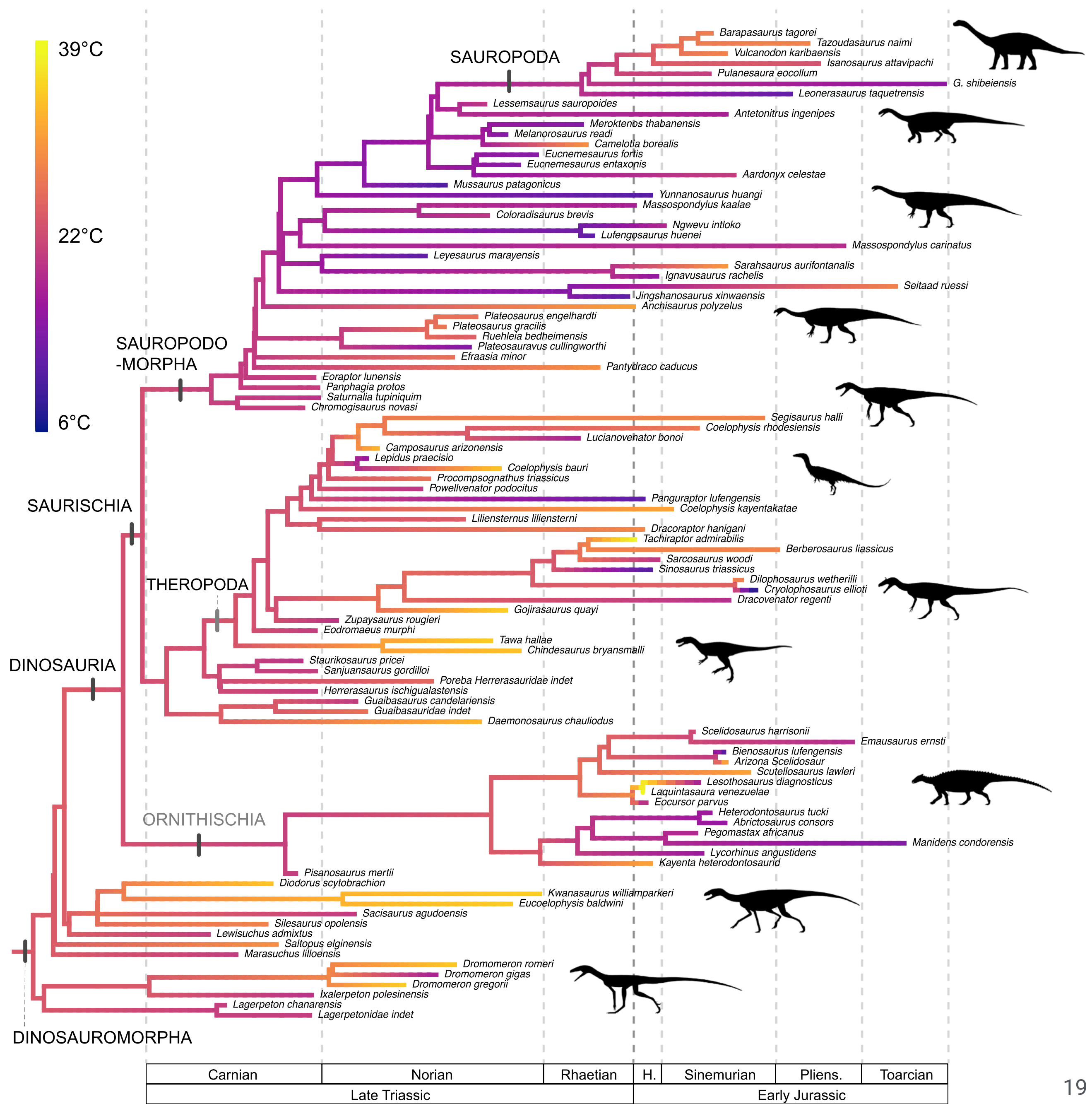
→ upwelling resulted in high prey densities

Example: niche evolution in sauropodomorphs

Traits are climate variables

Shifts to warmer niches early on

Provides evidence that climate influence evolution of the group



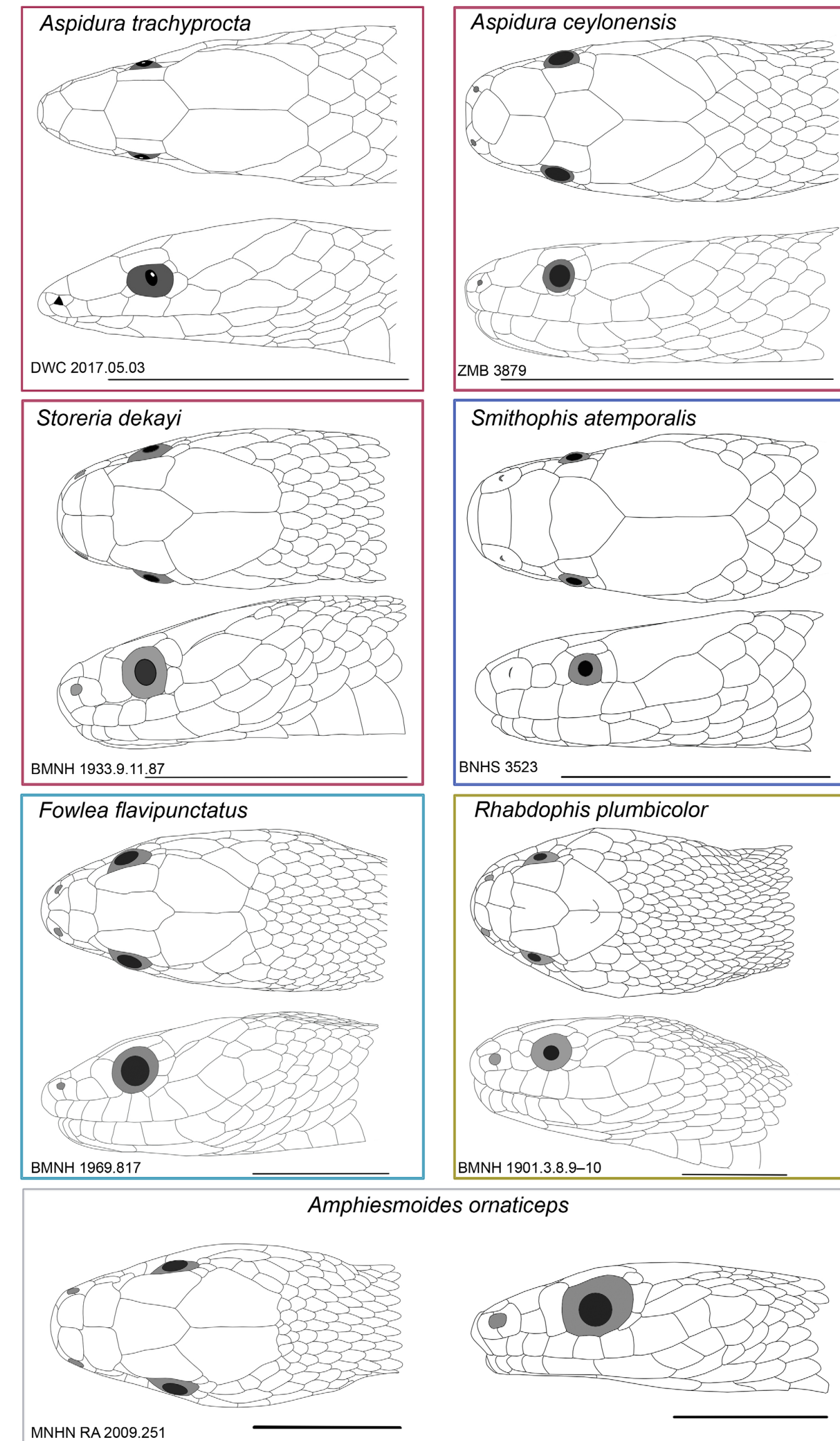
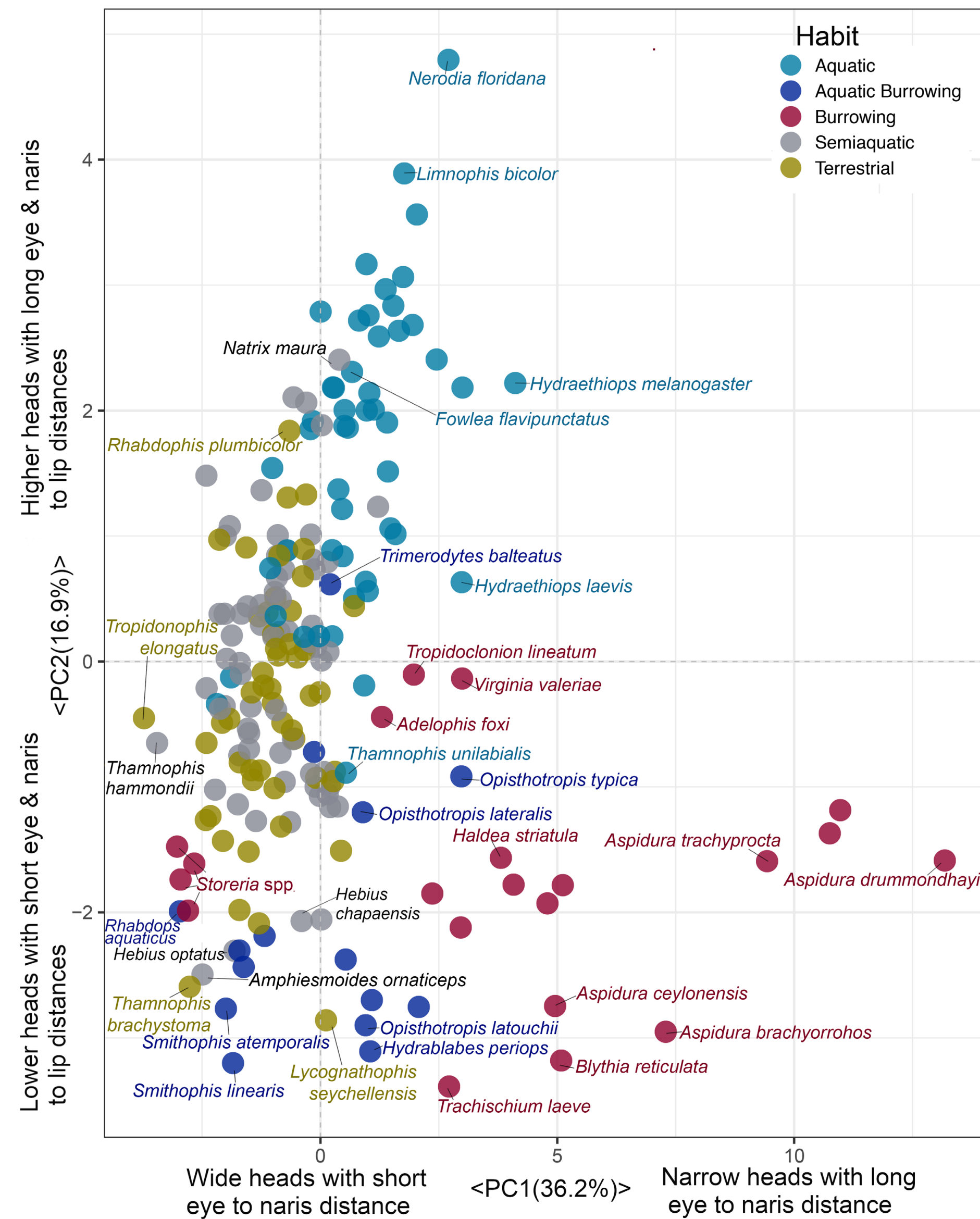


Exercise

Natricine snakes

Exercise by Nathalie Cooper

Data from *Deepak et al.* ([2022](#))



Further resources

Natalie Cooper has a comprehensive [online book](#) + accompanying [exercises](#)

Luke Harmon also has an excellent [online book](#)

Emma Dunne has a nice tutorial about [frog eyes](#) (phylogenetic signal, PGLS, continuous trait models, model selection)

Laura Soul has a nice tutorial about [feeding ecology in crinoids](#) (PGLS, continuous trait models, discrete models, model selection, model adequacy)